

File permissions in Linux

Project description

In this project, I worked with basic Linux commands to understand and manage file system permissions, which are critical for maintaining system security. I navigated directories using the `cd` command and examined detailed file and directory information with `ls -la` to analyze ownership and permission settings. I interpreted permission structures such as `drwxrwxrwx` to evaluate read, write, and execute access for the owner, group, and others. Additionally, I used the `chmod g+w filename` command to modify group permissions, demonstrating practical application of access control and the principle of least privilege. This project strengthened my understanding of Linux-based security fundamentals and system administration.

Check file and directory details

I used the `ls -la` command to view detailed information about files and directories, including hidden files. This command displays file type, permission settings, number of links, owner, group, file size, and last modification date. By reviewing these details, I was able to identify how access to files and directories is configured and assess potential security risks.

```
researcher2@b1120c1c161c:~/project$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:49 ..
-rw--w---- 1 researcher2 research_team  46 Jan  9 21:28 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 21:28 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jan  9 21:28 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 21:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_t.txt
```

Describe the permissions string

The permissions string (for example, `drwxrwxrwx`) represents access rights for a file or directory. The first character indicates the file type (`d` for directory, `-` for file). The remaining characters are grouped into three sets showing permissions for the owner, group, and others. Each set includes read (`r`), write (`w`), and execute (`x`) permissions. Understanding this string helps evaluate whether a resource is appropriately secured or overly permissive.

Change file permissions

I used the `chmod` command to modify file permissions and control user access. For example, the command `chmod g+w filename` grants write permission to the group while leaving other permissions unchanged. This demonstrates how Linux access controls can be adjusted to ensure users have only the permissions necessary to perform their tasks.

```
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:49 ..
-rw--w---- 1 researcher2 research_team  46 Jan  9 21:28 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 21:28 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jan  9 21:28 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 21:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_t.txt
researcher2@b1120cfc161c:~/projects$ chmod o-w project_k.txt
researcher2@b1120cfc161c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:49 ..
-rw--w---- 1 researcher2 research_team  46 Jan  9 21:28 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 21:28 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jan  9 21:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_t.txt
```

Change file permissions on a hidden file

Hidden files in Linux (files starting with a dot, such as `.config`) can also have their permissions modified using the same permission commands. I used `ls -la` to locate hidden files and applied the `chmod` command to change their permissions. This reinforced the understanding that hidden files are not more secure by default and must be properly protected using correct permission settings.

```

researcher2@b1120cfc161c:~/projects$ chmod u-w .project_x.txt
researcher2@b1120cfc161c:~/projects$ chmod g-w .project_x.txt
researcher2@b1120cfc161c:~/projects$ chmod g+r .project_x.txt
researcher2@b1120cfc161c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:49 ..
-r--r----- 1 researcher2 research_team  46 Jan  9 21:28 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 21:28 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_k.txt
-rw----- 1 researcher2 research_team  46 Jan  9 21:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_t.txt

```

Change directory permissions

I used the `chmod` command to modify directory permissions and control access to directories. By changing read, write, and execute permissions, I managed who could view directory contents, create or delete files, and access the directory itself. Understanding directory permissions helped me apply proper access control and prevent unauthorized users from modifying or accessing sensitive directories.

```

drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:49 ..
-r--r----- 1 researcher2 research_team  46 Jan  9 21:28 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jan  9 21:28 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_k.txt
-rw----- 1 researcher2 research_team  46 Jan  9 21:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_t.txt
researcher2@b1120cfc161c:~/projects$ chmod g-w drafts
researcher2@b1120cfc161c:~/projects$ chmod g-x drafts
researcher2@b1120cfc161c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:28 .
drwxr-xr-x 3 researcher2 research_team 4096 Jan  9 21:49 ..
-r--r----- 1 researcher2 research_team  46 Jan  9 21:28 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jan  9 21:28 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_k.txt
-rw----- 1 researcher2 research_team  46 Jan  9 21:28 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jan  9 21:28 project_t.txt

```

Summary

This project strengthened my understanding of Linux file and directory permissions and their role in system security. By using basic Linux commands such as `cd`, `ls -la`, and `chmod`, I learned how to inspect permission settings, interpret permission strings, and modify access for files, hidden files, and directories. These hands-on exercises improved my ability to apply the principle of least privilege and manage access controls in a Linux environment, which is a fundamental skill in cybersecurity.